

A Non-Log-Concave Obstruction to the Marginal Information-Production Inequality

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Status

Scoped counterexample, likely valid. This packet answers the narrow non-log-concave question around inequality (3.2) in Ball–Nguyen [1]. It does not refute their log-concave entropy-jump theorem or Lemma 4 under its stated positivity/log-concavity hypothesis.

Source Question

The source proves a dimension-free entropy jump for isotropic log-concave vectors. Immediately after Theorem 1, it says that log-concavity is crucial for the proof of inequality (3.2), and that the authors do not know whether that inequality holds without this assumption.

log-concave random vectors, i.e. those having a density f such that $-\log(f)$ is convex on $\mathbb{R}^n$. It is well known that such random vectors have finite entropy and have a positive spectral gap (see below). This family is central in many high-dimensional problems.

Theorem 1. *Let X be an isotropic log-concave random vector in $\mathbb{R}^d$. Assume its density f satisfies a Poincaré inequality (1.2) with constant $c > 0$. Then, if Y is an independent copy of X , we have*

$$\text{Ent}\left(\frac{X+Y}{\sqrt{2}}\right) - \text{Ent}(X) \geq \frac{c}{4(1+c)}(\text{Ent}(G) - \text{Ent}(X)). \quad (1.4)$$

Since $c \leq 1$ when f is isotropic, the constant $\frac{c}{4(1+c)}$ may be replaced by $\frac{c}{8}$. The log-concavity assumption will be crucial on the proof of inequality (3.2) below, we do not know whether it holds without this assumption.

As mentioned above, we need to develop a method different to the one of [3]. Our alternative approach relies on the study of *second* derivatives of the entropy along the heat semi-group (or rather along the Ornstein-Uhlenbeck semi-group). Note that it also gives

The inequality in question is the marginal information-production bound from Lemma 4. In the notation of the source, if h is the marginal of a smooth positive density $\omega = e^{-\phi}$ under the orthogonal projection onto E , the displayed bound is

$$\int_E \text{Tr}[(\text{Hess} \log h)^2] h \leq \int_{\mathbb{R}^N} \text{Tr}\left[(P_E(\text{Hess} \log \omega)P_E)^2\right] \omega. \quad (\text{BN})$$

the projection $P_E \omega$ of ω onto a subspace $E \subset \mathbb{R}^N$. It has a density on $E \simeq \mathbb{R}^{\dim E}$ which we denote by h . A useful observation due to Carlen [10] for Fisher information is that

$$J(h) \leq \int \frac{|P_E \nabla \omega|^2}{\omega}.$$

The next result provides an analogue for information production. However, we are able to establish it only in the case of log-concave densities: here is where the restriction in our main theorem comes from. Using it, we can then state the central inequality that will be used in the proof of the main theorem.

Lemma 4. *Let $N \geq 1$ and $\omega = e^{-\phi} : \mathbb{R}^N \rightarrow \mathbb{R}_+$ be a smooth positive function. Given a subspace $E \subset \mathbb{R}^N$ define the marginal function on E by*

$$\forall x \in E, \quad h(x) := e^{-\psi(x)} := \int_{E^\perp} \omega(x+y) dy = \int_{E^\perp} e^{-\phi(x+y)} dy.$$

Denote by P_E the orthogonal projection onto E . Then, for every $x \in E$ we have,

$$h(x) \text{Hess } \psi(x) \leq \int_{E^\perp} \omega(x+y) P_E \text{Hess } \phi(x+y) P_E dy$$

in the operator sense (for symmetric operators on E) and if $\text{Hess } \psi(x) \geq 0$, then

$$\text{Tr} \left[(\text{Hess } \psi(x))^2 h(x) \right] \leq \int_{E^\perp} \text{Tr} \left[(P_E \text{Hess } \phi(x+y) P_E)^2 \right] \omega(x+y) dy. \quad (3.1)$$

Therefore, if $\text{Hess } \psi \geq 0$ we have

$$\int_E \text{Tr} \left[(\text{Hess } \log h)^2 \right] h \leq \int_{\mathbb{R}^N} \text{Tr} \left[(P_E (\text{Hess } \log \omega) P_E)^2 \right] \omega. \quad (3.2)$$

Proof. We start with the observation that for $x \in E$,

$$\frac{\nabla h(x) \otimes \nabla h(x)}{h(x)} \leq \int \frac{P_E \nabla \omega(x+y) \otimes P_E \nabla \omega(x+y)}{\omega(x+y)} dy, \quad (3.3)$$

Idea of the Proof

In one dimension, (BN) compares the squared curvature of the log of a marginal density with the averaged squared x -curvature of the joint log-density. If $X | Y = y$ is always a unit Gaussian with mean depending on y , then $\partial_{xx} \log \omega = -1$ everywhere, so the right side is exactly 1. But if the conditional means are concentrated near two well-separated values, the X -marginal is a two-bump Gaussian mixture. Near the saddle between the two bumps, $\log h$ has large positive curvature. For a suitable separation, the squared curvature integral exceeds the constant 1.

Counterexample

Theorem. *The inequality (BN) is false without the log-concavity/sign mechanism used in Lemma 4 of [1]. More precisely, there is a smooth strictly positive probability density $\omega : \mathbb{R}^2 \rightarrow (0, \infty)$ such that, for the projection $(x, y) \mapsto x$,*

$$\int_{\mathbb{R}} ((\log h)''(x))^2 h(x) dx > \int_{\mathbb{R}^2} (\partial_{xx} \log \omega(x, y))^2 \omega(x, y) dx dy,$$

where $h(x) = \int_{\mathbb{R}} \omega(x, y) dy$.

Proof. Let $a = 5/2$. First consider the limiting one-dimensional mixture

$$h_0(x) = \frac{1}{2}\gamma(x-a) + \frac{1}{2}\gamma(x+a), \quad \gamma(x) = (2\pi)^{-1/2}e^{-x^2/2}.$$

Equivalently,

$$h_0(x) = (2\pi)^{-1/2}e^{-(x^2+a^2)/2} \cosh(ax).$$

Thus

$$(\log h_0)''(x) = -1 + a^2 \operatorname{sech}^2(ax).$$

Let

$$L_0 = \int_{\mathbb{R}} ((\log h_0)''(x))^2 h_0(x) dx.$$

A direct expansion gives

$$L_0 - 1 = \frac{a^2 e^{-a^2/2}}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-x^2/2} \operatorname{sech}(ax) (a^2 \operatorname{sech}^2(ax) - 2) dx. \quad (1)$$

We show that the integral in (1) is positive for $a = 5/2$.

On $|x| \leq 0.3$, we have $a|x| \leq 0.75$, hence

$$e^{-x^2/2} \operatorname{sech}(ax) (a^2 \operatorname{sech}^2(ax) - 2) \geq e^{-0.045} \operatorname{sech}(0.75) \left(\frac{25}{4} \operatorname{sech}^2(0.75) - 2 \right).$$

Therefore the contribution of $[-0.3, 0.3]$ is larger than

$$0.6 e^{-0.045} \operatorname{sech}(0.75) \left(\frac{25}{4} \operatorname{sech}^2(0.75) - 2 \right) > 0.765. \quad (2)$$

The integrand is negative only when

$$\operatorname{sech}(ax) < \frac{\sqrt{2}}{a},$$

that is, outside

$$|x| < \frac{1}{a} \operatorname{arcosh}(a/\sqrt{2}) > 0.46.$$

On that negative region,

$$|\operatorname{sech}(ax)(a^2 \operatorname{sech}^2(ax) - 2)| \leq 2 \operatorname{sech}(ax) \leq 4e^{-a|x|}.$$

Consequently the total negative contribution is bounded in magnitude by

$$8 \int_{0.46}^{\infty} e^{-x^2/2 - (5/2)x} dx = 8e^{25/8} \sqrt{\frac{\pi}{2}} \operatorname{erfc}\left(\frac{2.96}{\sqrt{2}}\right) < 0.703. \quad (3)$$

The lower bound (2) and upper bound (3) prove that the integral in (1) is positive. Hence $L_0 > 1$.

Now smooth the two-point mixer. Let Y_σ have density

$$p_\sigma(y) = \frac{1}{2} \frac{1}{\sigma} \gamma\left(\frac{y-1}{\sigma}\right) + \frac{1}{2} \frac{1}{\sigma} \gamma\left(\frac{y+1}{\sigma}\right),$$

and, conditionally on $Y_\sigma = y$, let X_σ be Gaussian with mean ay and variance 1. The joint density is

$$\omega_\sigma(x, y) = p_\sigma(y) \gamma(x - ay),$$

which is smooth and strictly positive on $\mathbb{R}^2$. Since

$$\partial_{xx} \log \omega_\sigma(x, y) = -1$$

everywhere, the right side of (BN) for the projection onto the x -axis is exactly

$$\int_{\mathbb{R}^2} 1 \cdot \omega_\sigma(x, y) dx dy = 1.$$

The x -marginal of ω_σ is

$$h_\sigma(x) = \frac{1}{2} \gamma_s(x - a) + \frac{1}{2} \gamma_s(x + a), \quad s^2 = 1 + a^2 \sigma^2,$$

where γ_s is the centered Gaussian density of variance s^2 . As $\sigma \downarrow 0$, h_σ and the first two derivatives of $\log h_\sigma$ converge pointwise to the corresponding quantities for h_0 ; on $0 \leq \sigma \leq \sigma_0$ they are dominated by an integrable Gaussian-tail bound. Dominated convergence gives

$$\int_{\mathbb{R}} ((\log h_\sigma)''(x))^2 h_\sigma(x) dx \rightarrow L_0 > 1.$$

Thus, for all sufficiently small positive σ , the left side of (BN) is larger than 1, while the right side is exactly 1. This proves the claimed smooth positive-density counterexample. $\square$

Verification Notes

The proof above is analytic. The script `code/mixture_curvature_check.py` only checks the numerical size of the gap for finite smoothing. For $a = 5/2$, it prints values such as

$$L_{0.05} - 1 \approx 0.143185 > 0.$$

This supports the continuity step but is not used as a proof.

Limitations

This packet refutes only the log-concavity-free version of (BN). It leaves intact:

- the source theorem for isotropic log-concave random vectors;
- Lemma 4 under the stated $\text{Hess } \psi \geq 0$ sign condition;
- any non-log-concave variant that imposes a replacement sign condition on the marginal potential $\psi = -\log h$.

Novelty and Search Bounds

Local cheap indexes were searched for arXiv:1206.5098, entropy jumps, information production, log-concavity, marginal Hessian inequalities, KLS, thin shell, and slicing/hyperplane. Existing run memory already covers the later resolution of slicing/hyperplane and thin-shell conjecture targets, but no packet covered this non-log-concave marginal-information inequality.

Web searches on 2026-07-18 for the source title, the phrase “we do not know whether”, “marg1bis”, “information production marginal log-concave”, and nearby entropy-jump literature found the source paper and later log-concave extensions such as Bizeul [2], but no explicit counterexample to the non-log-concave form of (BN).

Human Review Recommendation

Accept as a scoped counterexample to removing log-concavity/sign assumptions from the marginal information-production inequality (3.2). Do not count it as a counterexample to the source's log-concave entropy-jump theorem.

References

- [1] K. Ball and V. H. Nguyen, *Entropy jumps for isotropic log-concave random vectors and spectral gap*, Studia Math. 213 (2012), no. 1, 81–96. arXiv:1206.5098.
- [2] P. Bizeul, *Entropy and Information jump for log-concave vectors*, Comptes Rendus. Mathématique 361 (2023), 487–493.